5.1 Attention cues

- information in our environment is not scarce, attention is
- when inspecting a visual scene, our optic nerve receives information at the order of 10^8 bits per second, far exceeding what our brain can fully process
- fortunately, we learned from experience (also known as data) that *not all sensory inputs are equal*
- throughout human history, the capability of directing attention to only a fraction of information of interest has enabled our brain to allocate resources more smartly to survive, to grow, and to socialize, such as detecting predators, preys, and mates

5.1 Attention cues

- to explain how our attention is deployed in the visual world, a two-component framework has emerged
- in this framework, subjects selectively direct the spotlight of attention using both the *nonvolitional cue* and the *volitional cue*
- the nonvolitional cue is based on the *saliency* of objects in the environment
- imagine there are five objects in front of us: a newspaper, a research paper, a cup of coffee, a notebook, and a book, such as in Figure 1
- while all the paper products are printed in black and white, the coffee cup is red
- in other words, this coffee is intrinsically salient in this visual environment, automatically and involuntarily drawing attention
- so, we bring the fovea (the center of the macula, where visual acuity is highest) onto the coffee, as shown in Figure 1

5.1 Attention cues

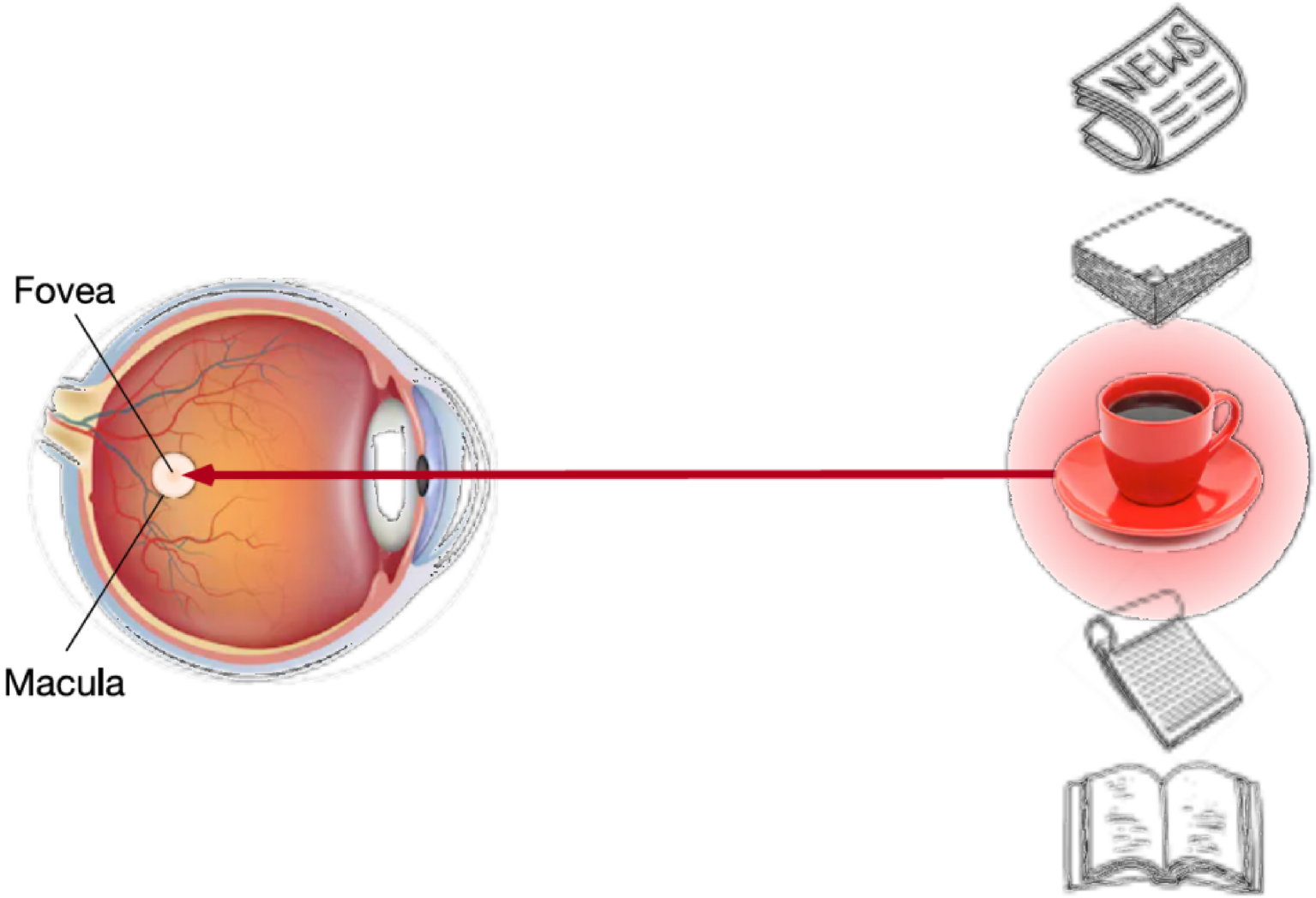


Figure 1: Using the nonvolitional cue based on saliency (red cup, non-paper), attention is involuntarily directed to the coffee.

5.1 Attention cues

- after drinking coffee, we become caffeinated, and want to read a book
- so we turn our head, refocus our eyes, and look at the book, as depicted in Figure 2
- different from the case in Figure 1, where the coffee biases us towards selecting based on saliency, in this task-dependent case, we select the book under cognitive and volitional control
- using the volitional cue based on variable selection criteria, this form of attention is more deliberate
- it is also more powerful, with the subject's voluntary effort

5.1 Attention cues

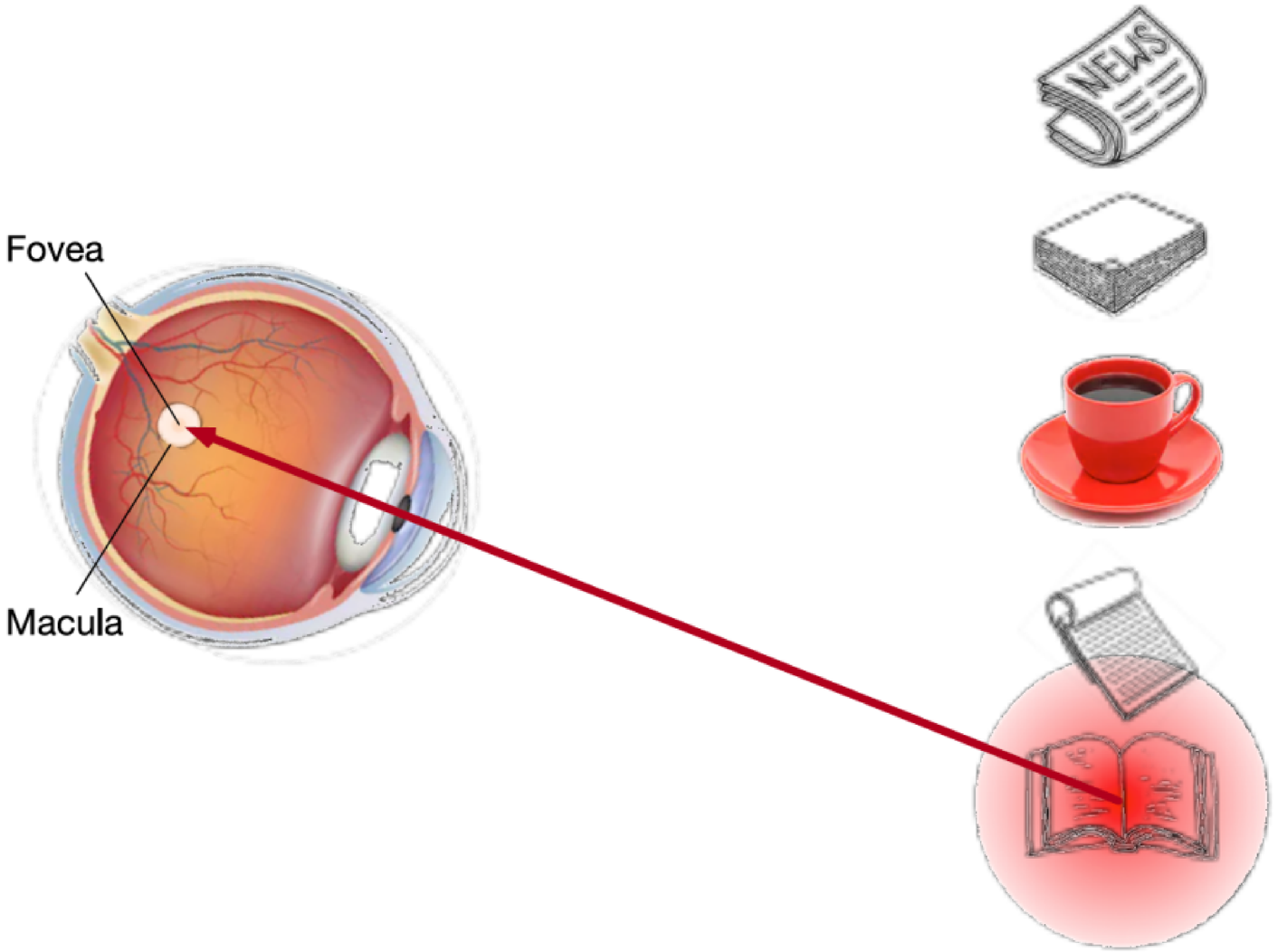


Figure 2: Using the volitional cue (want to read a book) that is task-dependent, attention is directed to the book under volitional control.

5.1 Attention cues

- inspired by the nonvolitional and volitional attention cues that explain the attentional deployment, in the following we will describe a framework for designing attention mechanisms by incorporating these two attention cues
- to begin with, consider the simpler case where only nonvolitional cues are available
- to bias selection over sensory inputs, we can simply use a parameterized fully-connected layer or even a non-parameterized max or average pooling layer

5.1 Attention cues

- therefore, what sets attention mechanisms apart from those fully-connected layers or pooling layers is the inclusion of the volitional cues
- in the context of attention mechanisms, we refer to volitional cues as *queries*
- given any query, attention mechanisms bias selection over sensory inputs (e.g., intermediate feature representations) via *attention pooling*
- these sensory inputs are called *values*, in the context of attention mechanisms
- more generally, every value is paired with a *key*, which can be thought of as the nonvolitional cue of that sensory input
- as shown in Figure 3, we can design attention pooling so that the given query (volitional cue) can interact with keys (nonvolitional cues), which guides bias selection over values (sensory inputs)

5.1 Attention cues

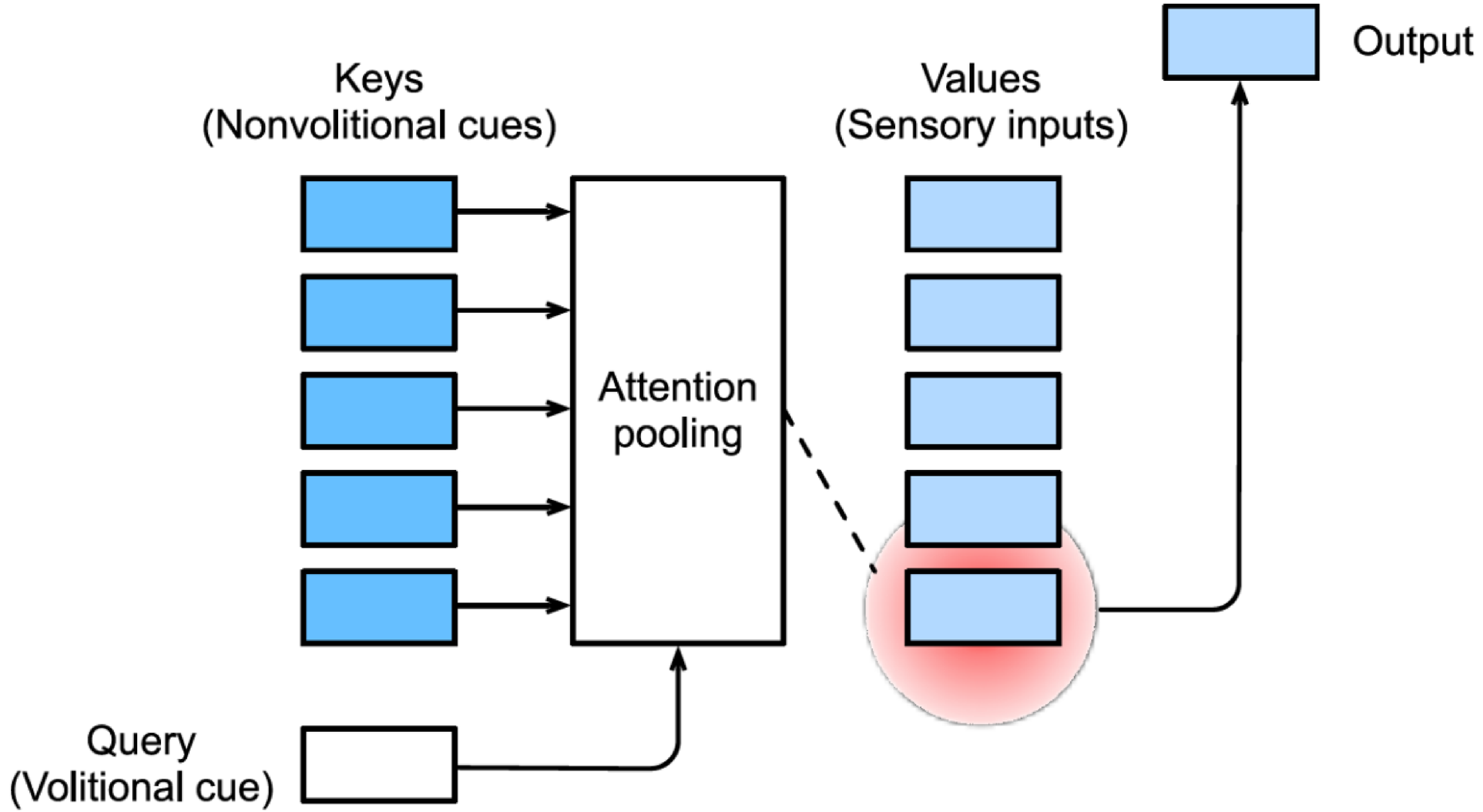


Figure 3: Attention mechanisms bias selection over values (sensory inputs) via attention pooling, which incorporates queries (volitional cues) and keys (nonvolitional cues).